Multiple Sequence Alignment of Cytochrome c Protein Using Clustal Omega

OBJECTIVE:

The objective of this task is to perform multiple sequence alignment (MSA) of Cytochrome c protein sequences obtained from different species using Clustal Omega. The purpose of this analysis is to compare the amino acid sequences and identify conserved residues that are functionally or structurally important. By aligning multiple sequences from the same protein family, the study aims to evaluate sequence similarity, detect conserved motifs, and observe variations among species. This analysis also helps in understanding evolutionary relationships and determining the degree of conservation of the Cytochrome c protein across different organisms.

INTRODUCTION:

Multiple Sequence Alignment (MSA) is an essential bioinformatics technique used to align three or more biological sequences, such as proteins or DNA, to identify similarities and differences among them. It helps in detecting conserved regions, functional motifs, and structural domains that are important for biological activity. Conserved residues often indicate evolutionary stability and functional significance, while variations may reflect species-specific adaptations.

Cytochrome c is a small heme-containing protein that plays a critical role in the electron transport chain during cellular respiration. It is involved in the transfer of electrons between Complex III and Complex IV in mitochondria, contributing to ATP production. Cytochrome c is highly conserved across diverse organisms ranging from mammals to birds and other vertebrates.Analyzing multiple Cytochrome c sequences from different species allows the identification of conserved amino acids that are essential for its structure and function. In this study, Cytochrome c protein sequences from selected organisms were retrieved and aligned using

Clustal Omega, and the resulting alignment was examined to evaluate sequence conservation and similarity patterns.

METHODOLOGY:

1. **Sequence Retrieval**

   Cytochrome c protein sequences from five different species were obtained from the National Center for Biotechnology Information (NCBI) protein database. The selected sequences belong to the same protein family to ensure meaningful comparison. Each sequence was downloaded in FASTA format.

2. **Preparation of Input File**

   The retrieved FASTA sequences were combined into a single text file. Each sequence was properly labeled with its organism name or accession number to ensure correct identification during alignment.

3. **Multiple Sequence Alignment**

   The combined FASTA file was submitted to the Clustal Omega tool available through the European Bioinformatics Institute (EBI) web server. Default parameters were used to perform the multiple sequence alignment.

4. **Result Analysis**

   The alignment output was examined to identify conserved regions and sequence variations. Conserved amino acids were indicated by (*) symbols, while (:) and (.) represented strong and weak similarities, respectively. Gaps (-) introduced during alignment were also observed.

5. **Phylogenetic Analysis**

   A guide tree (phylogenetic tree) generated by Clustal Omega was analyzed to understand the evolutionary relationships among the selected sequences.

RESULTS AND INTERPRETATION:

Multiple sequence alignment of five Cytochrome c protein sequences was successfully performed using Clustal Omega. The alignment output displayed the arranged amino acid sequences along with symbols indicating the level of conservation among the sequences.

A significant number of conserved residues were observed, represented by the (*) symbol in the alignment. These fully conserved positions indicate identical amino acids across all selected species, suggesting strong evolutionary conservation of Cytochrome c. Several positions also showed (:) and (.) symbols, representing strong and weak similarities, respectively. These similarities indicate conservative substitutions where amino acids with similar biochemical properties are present.A few gaps (-) were introduced in certain regions to optimize the alignment. These gaps represent minor variations or insertions/deletions that occurred during evolution. A characteristic conserved motif of the Cytochrome c family, the **CXXCH heme-binding motif**, was observed, confirming that all selected sequences belong to the same protein family and share a common function in electron transport.

The guide tree generated from the alignment illustrated the evolutionary relationships among the selected species. Closely related sequences were grouped together with shorter branch lengths, while more distantly related species showed longer branches, indicating greater evolutionary divergence.Overall, the alignment results demonstrate that Cytochrome c is highly conserved across species, reflecting its essential role in cellular respiration and energy production.

CONCLUSION:

The multiple sequence alignment of Cytochrome c proteins from different species revealed a high level of sequence conservation, indicating the functional and structural importance of this protein. The presence of numerous conserved residues across all sequences suggests that Cytochrome c has been evolutionarily preserved due to its critical role in cellular respiration and energy production. Minor variations and gaps observed in some regions reflect evolutionary divergence among species. The phylogenetic analysis further supported the evolutionary relationships, grouping closely related organisms together. Overall, the study confirms that Cytochrome c is a highly conserved protein with significant biological importance across different species.

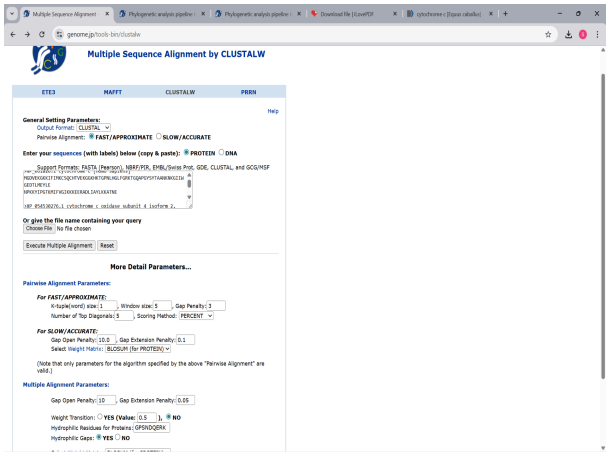
```
>NP_061820.1 cytochrome c [Homo sapiens]
MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIWGEDTLMEYLE
NPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE

>XP_054530276.1 cytochrome c oxidase subunit 4 isoform 2, mitochondrial isoform X1 [Pan troglodytes]
MLPRAAWSLVLRKGGGGRRGMHSSEGTTHGGGKMSPYTNCYAQRYYPMPEEPFCTELNAEQQALKEKEKG
SWTQLTHAEKVALYRLQFNETFAEMNRRSNEWKTVMGCVFFFIGFAALVIWWQRVYVFPPKPITLTDEWK
AQQLQRMLDMKVNPVQGLASRWDYEKKQWKK

>AEP27246.1 cytochrome c [Mus musculus]
MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANKNKGITWGEDTLMEYLE
NPKKYIPGTKMIFAGIKKKGERADLIAYLKKATNE

>AHN54498.1 cytochrome c proximal, isoform B [Drosophila melanogaster]
MGVPAGDVEKGKKLFVQRCAQCHTVEAGGKHKVGPNLHGLIGRKTGQAAGFAYTDANKAKGITWNEDTLF
EYLENPKKYIPGTKMIFAGLKKPNERGDLIAYLKSATK

>NP_001157486.2 cytochrome c [Equus caballus]
MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGFTYTDANKNKGITWKEETLMEYLE
NPKKYIPGTKMIFAGIKKKTEREDLIAYLKKATNE
```



CLUSTALW Result

[clustalw.aln][clustalw.dnd][readme]

Select tree menu | Exec

```
 CLUSTAL 2.1 Multiple Sequence Alignments

Sequence type explicitly set to Protein
Sequence format is Pearson
Sequence 1: NP_061820.1        105 aa
Sequence 2: XP_054530276.1     171 aa
Sequence 3: AEP27246.1         105 aa
Sequence 4: AHN54498.1         108 aa
Sequence 5: NP_001157486.2     105 aa
Start of Pairwise alignments
Aligning...


Sequences (1:2) Aligned. Score: 12.381
Sequences (1:3) Aligned. Score: 91.4286
Sequences (1:4) Aligned. Score: 76.1905
Sequences (1:5) Aligned. Score: 88.5714
Sequences (2:3) Aligned. Score: 11.4286
Sequences (2:4) Aligned. Score: 12.037
Sequences (2:5) Aligned. Score: 13.3333
Sequences (3:4) Aligned. Score: 83.8095
Sequences (3:5) Aligned. Score: 94.2857
Sequences (4:5) Aligned. Score: 81.9048
Guide tree file created:    [clustalw.dnd]

There are 4 groups
Start of Multiple Alignment

Aligning...
Group 1: Sequences:    2       Score:1671
Group 2: Sequences:    3       Score:1668
Group 3: Sequences:    4       Score:1564
Group 4:                       Delayed
Alignment Score 3305

CLUSTAL-Alignment file created  [clustalw.aln]
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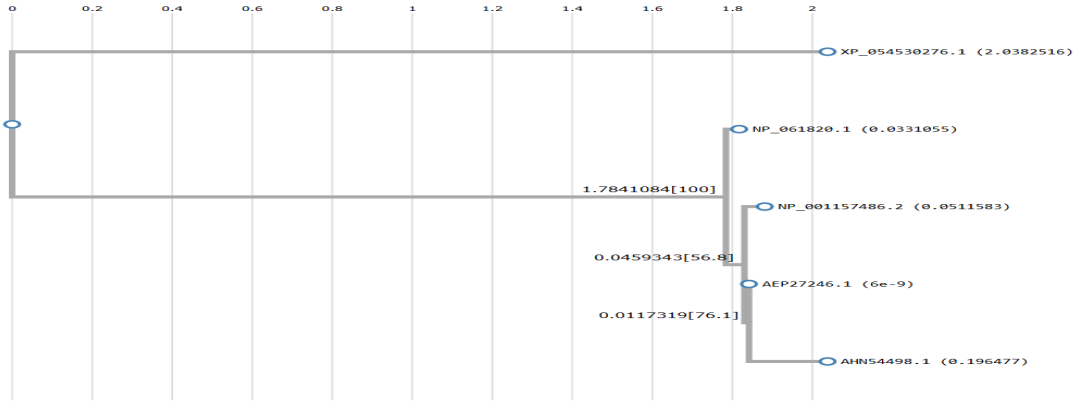
clustalw.dnd

```
(
(
NP_061820.1:0.06548,
AEP27246.1:0.02024)
:0.00833,
(
XP_054530276.1:0.78108,
AHN54498.1:0.09854)
:0.04947,
NP_001157486.2:0.03452);
```



clustalw.aln

```
CLUSTAL 2.1 multiple sequence alignment


NP_061820.1         -------MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAAN
AEP27246.1          -------MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDAN
NP_001157486.2      -------MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGFTYTDAN
AHN54498.1          ---MGVPAGDVEKGKKLFVQRCAQCHTVEAGGKHKVGPNLHGLIGRKTGQAAGFAYTDAN
XP_054530276.1      MLPRAAWSLVLRKGGGGRRGMHSSEGTTHGGGKMSPYTNCYAQRYYPMPEEPFCTELNAE
                              :.**        :.  *.. *** .  .* :.       : .  :   *:

NP_061820.1         KNKG-----IIWGEDTLMEYLE-----------------NPKKYIPGTKMIFVGIKKKEE
AEP27246.1          KNKG-----ITWGEDTLMEYLE-----------------NPKKYIPGTKMIFAGIKKKGE
NP_001157486.2      KNKG-----ITWKEETLMEYLE-----------------NPKKYIPGTKMIFAGIKKKTE
AHN54498.1          KAKG-----ITWNEDTLFEYLE-----------------NPKKYIPGTKMIFAGLKKPNE
XP_054530276.1      QQALKEKEKGSWTQLTHAEKVALYRLQFNETFAEMNRRSNEWKTVMGCVFFFIGFAALVI
                    :          * : *  * :                  *  * : *  ::* *:

NP_061820.1         RADLIAYLKKATNE-------------------------------------
AEP27246.1          RADLIAYLKKATNE-------------------------------------
NP_001157486.2      REDLIAYLKKATNE-------------------------------------
AHN54498.1          RGDLIAYLKSATK--------------------------------------
XP_054530276.1      WWQRVYVFPPKPITLTDEWKAQQLQRMLDMKVNPVQGLASRWDYEKKQWKK
                      : :  :   .
```

*** = fully conserved amino acids**

: or . = similar amino acids